



Representing Hard Problems As Collections Of Easy Ones?

Inspired by Nature?



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We are going to consider some phenomena from real engineering that can be represented in matrix form and note a particular type of behaviour

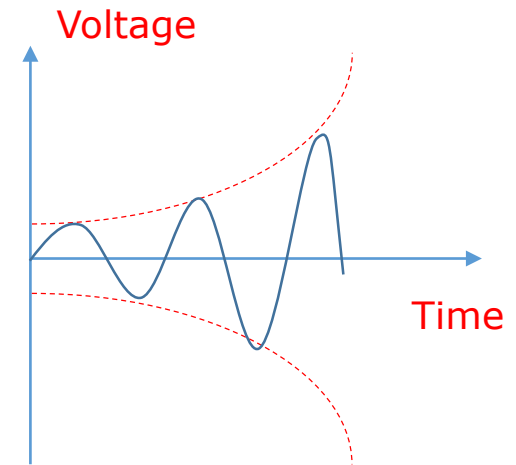
We will explore how the maths can explain this and then realise that nature has inspired us to discover an incredibly useful approach to problem solving

Feedback

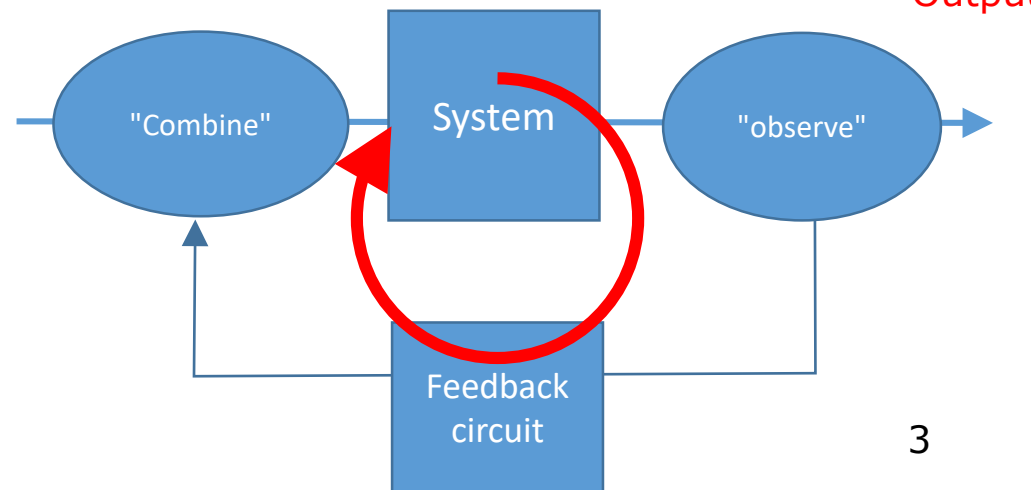
Why is guitar *feedback* a pure note?

We understand that there are *certain characteristic time waveforms* (sinusoids of particular frequencies) that just keep growing in magnitude as they "repeatedly go round the feedback loop"

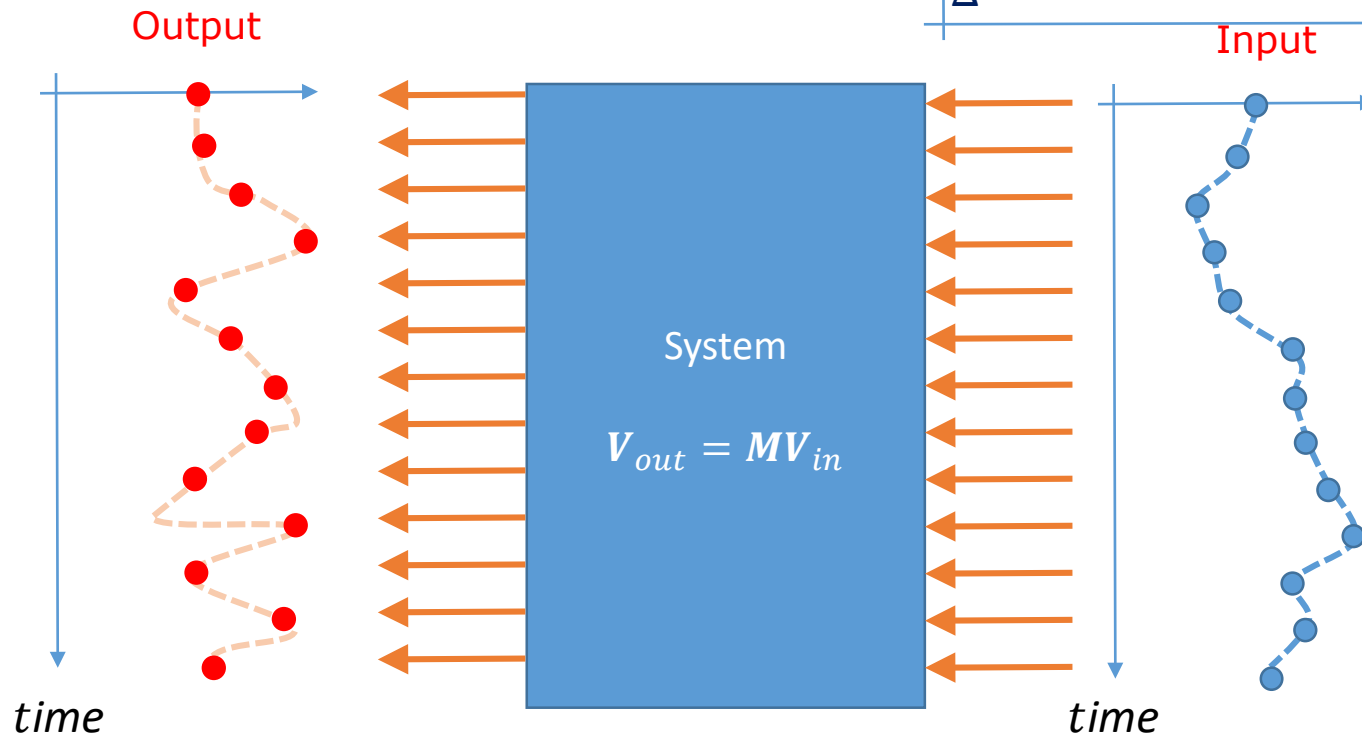
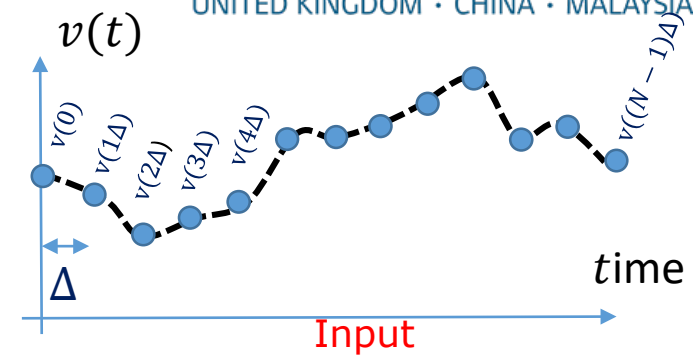
Let us introduce a matrix model of this



Input

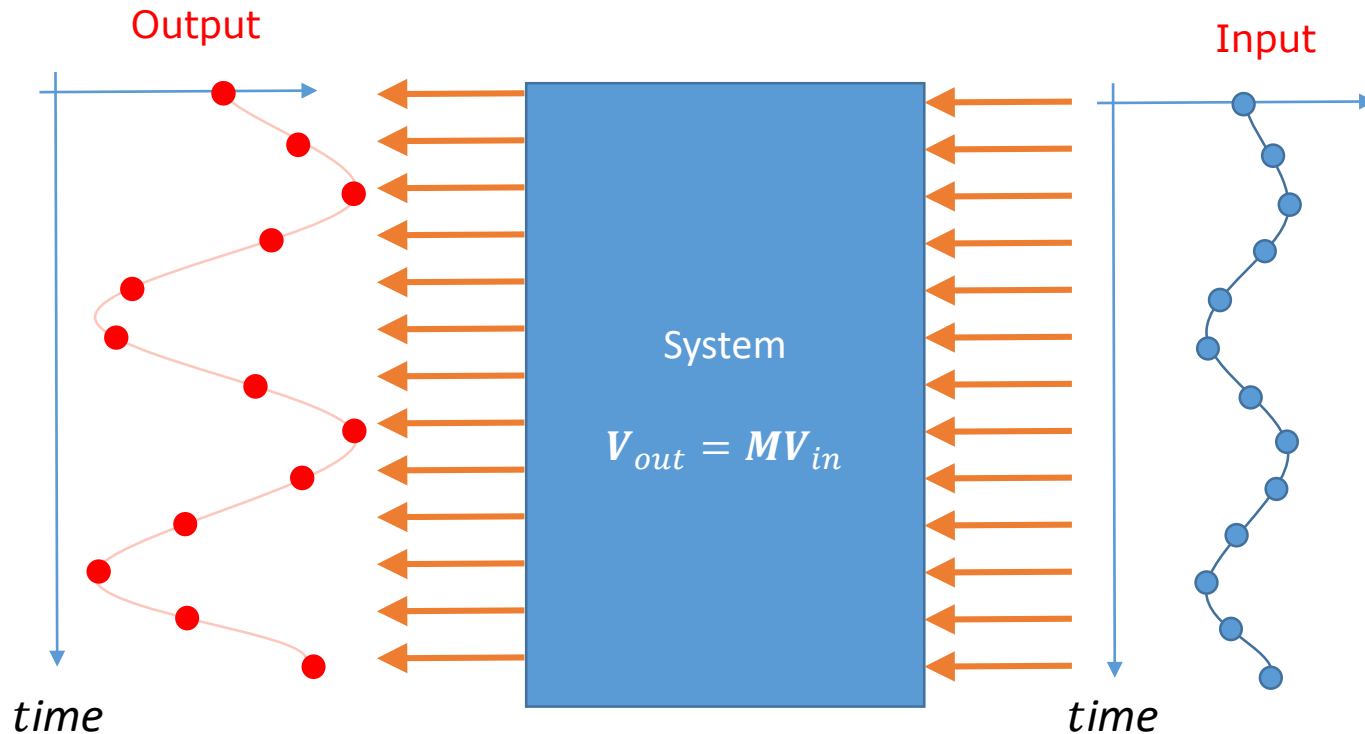


We will uniformly sample a waveform in time and regard each value as an *input to a system*



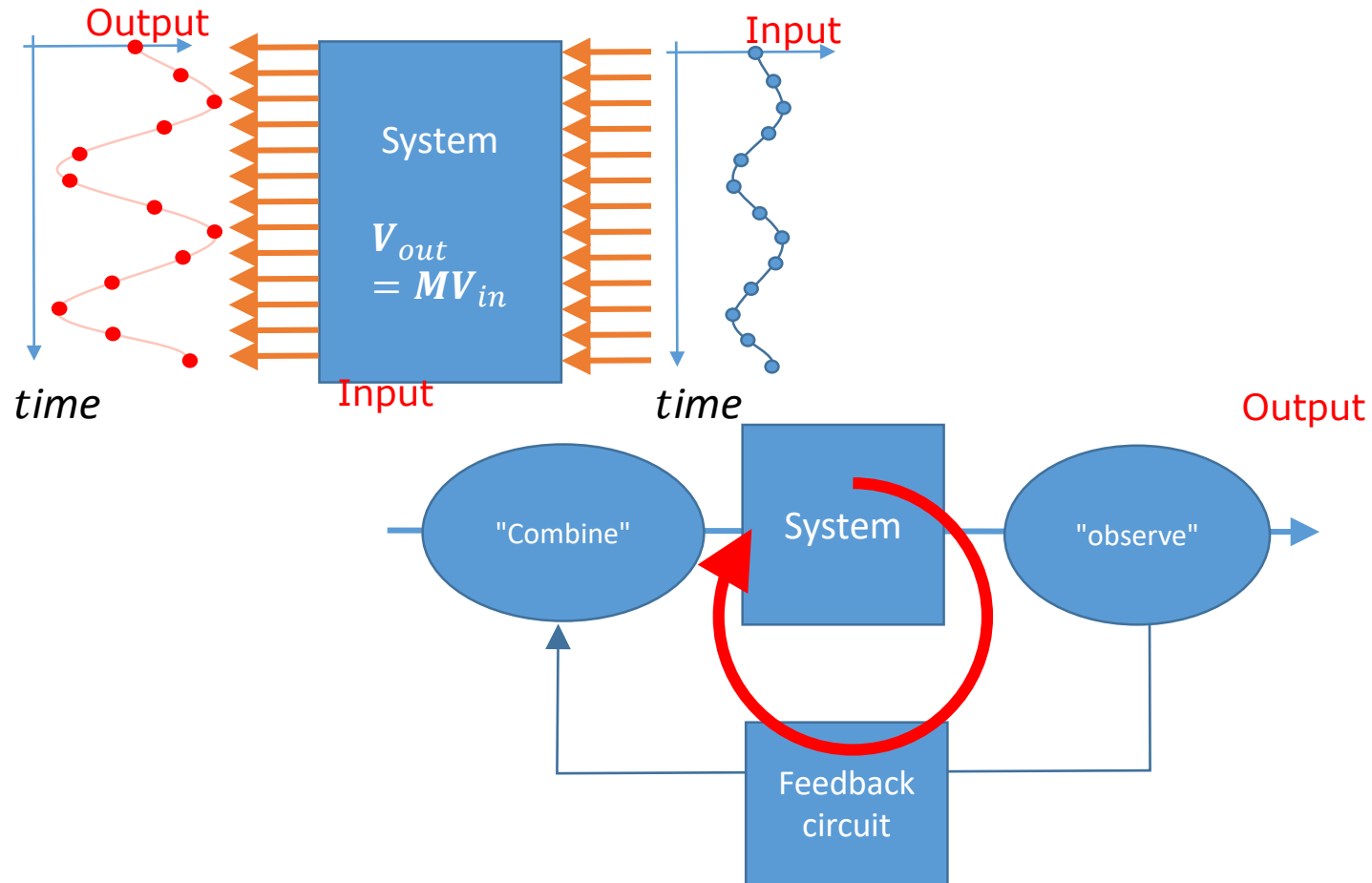
For an arbitrary set of input values, the system generates a different set of output values - we get a different "shape" waveform

However if the input waveform is a particular shape



Here, the effect of the system is simply that the waveform changes its magnitude, not its shape

Why does this explain feedback - because we then "connect the output to the input"



Only certain waveforms are purely amplified around the loop

If for this particular input the output is say k times larger than the input, then the system is behaving as a simple *amplifier*

For a general input

$$V_{out} = \mathbf{M}V_{in}$$

For the "special input"

$$V_{out} = k V_{in}$$

First of all this behaviour might appeal because we realise that simple scalar multiplication by k to get V_{out} from V_{in} is a lot "easier" than a full matrix multiplication (e.g. in a computer program)

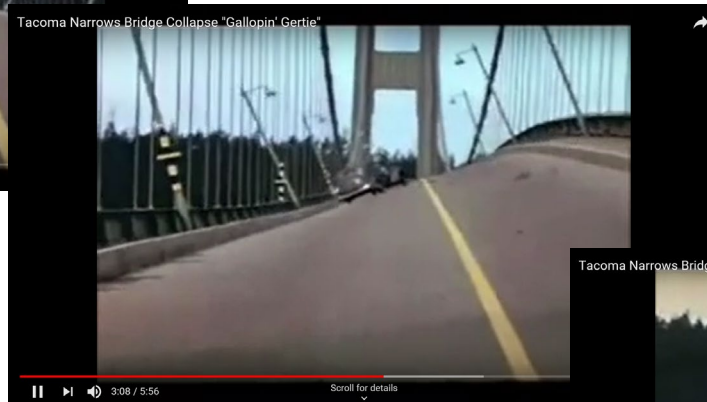
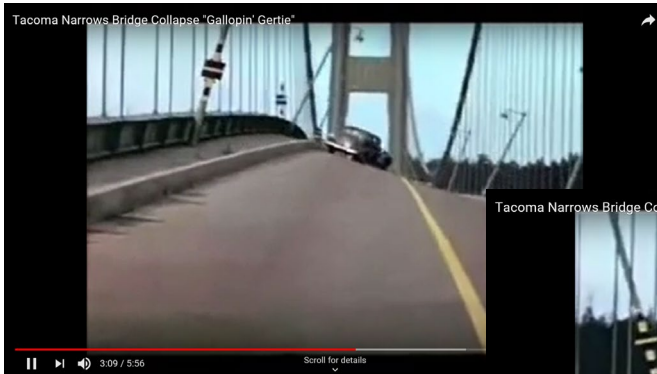
Second, such behaviour might be downright dangerous!

Millennium Bridge, London, 2000,

<https://www.youtube.com/watch?v=2cuXbpXRvJ0>

Tacoma Narrows Bridge, 1940,

<https://www.youtube.com/watch?v=j-zczJXSxmw>



If only the engineers had asked themselves;

*Are there any "inputs" such that the system
behaves as an ideal amplifier with feedback, i.e.*

$$\mathbf{M} \mathbf{V}_{in} = k \mathbf{V}_{in}$$

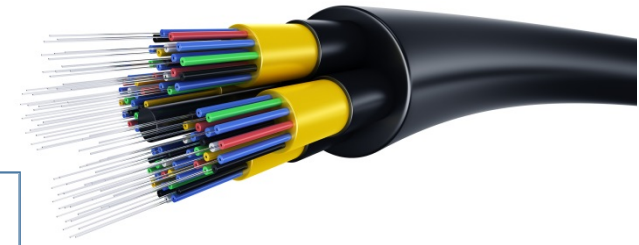
*because knowing if and when this is possible allows us to
protect against catastrophe*

Wave Propagation

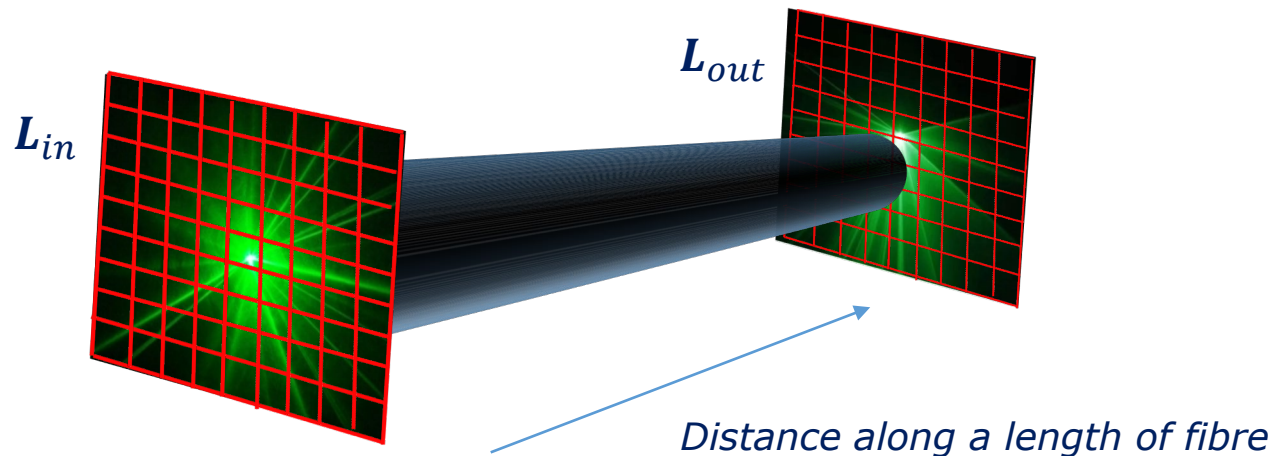


It's not all doom and gloom if " $\mathbf{M} \mathbf{V}_{in} = k \mathbf{V}_{in}$ "

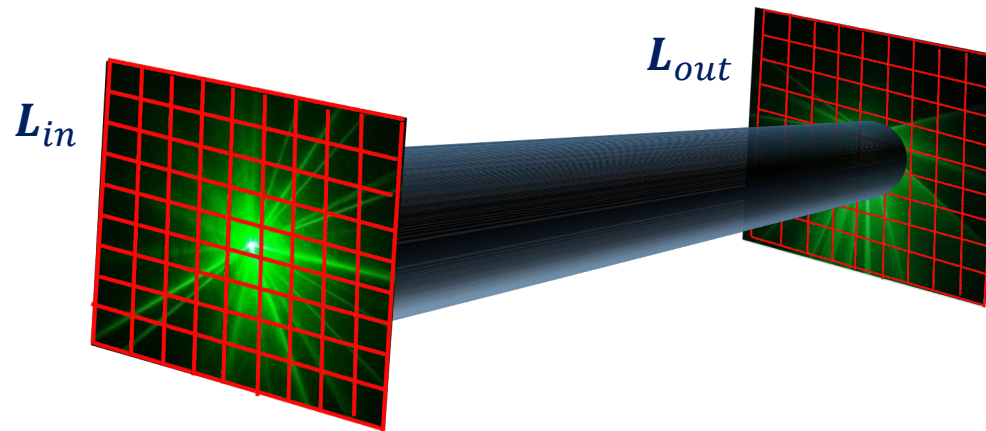
We would have no internet without this phenomenon



Here we are happy to think of an optical fibre as very fine glass tube. At each cross section we sample the "strength of the light" on a 2D grid (it's OK to think of voltage instead) and put these values in column vectors such as $\mathbf{L}_{in}$



Don't get confused, thinking that as we sampled on a 2D grid $\mathbf{L}_{in}$ is therefore a 2D matrix; No, we just list all the values sequentially in a 1D column vector



Physicists tell us that the relationship between the output samples and the input samples is expressible using a matrix

$$L_{out} = PL_{in}$$

If we know the light "pattern" at the start of the fibre we can evaluate the light pattern at the end of the fibre

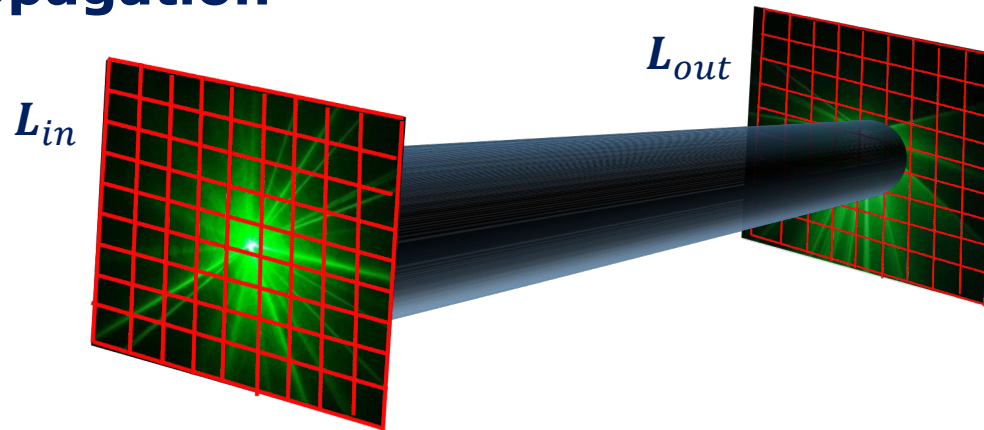
Generally, the light pattern is "all messed up" - changed a lot - by the fibre and it is really difficult to use it for data communications

Wave Propagation



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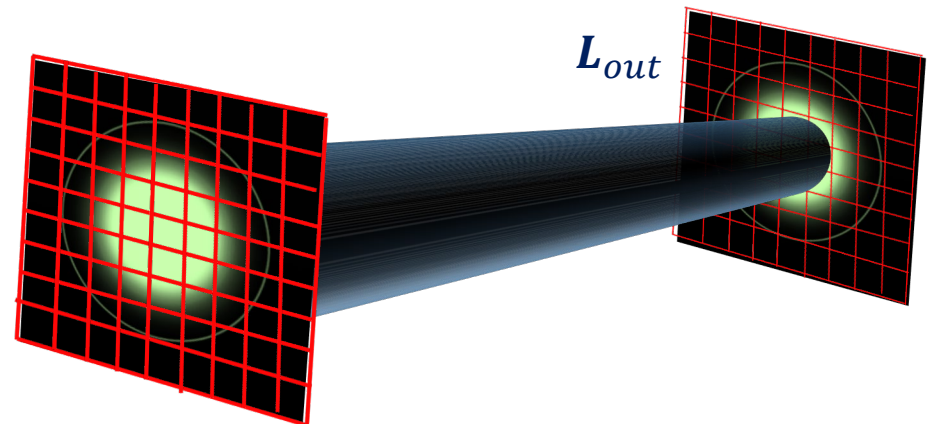
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What we need for high speed communications is to find a "nice light pattern" that when used as the input to the fibre, gives *the same shape pattern at the output - but maybe smaller in amplitude due to energy absorption in the glass*

So given that for a general input L_{in}

$$L_{out} = PL_{in}$$



We are seeking a particular input so that $L_{out} = kL_{in}$, hence

$$PL_{in} = kL_{in}$$



Inspired by Nature?

- and making £ as engineers

The common feature of the three examples, each of which can be expressed in matrix form, is that the output is proportional to the input $Ax = kx$

- Feedback - evolution of a waveform in time - stability - control - catastrophic "amplifications" - aerospace flight dynamics ...
- Propagation - evolution of a "cross-section shape" with distance - critical phenomenon for a range of communications topics - optical networks, microwave systems (e.g. mobiles), RADAR, ...
- and as we shall see, such "special solutions" lead to massive simplification of a range of computational problems and
- Provide engineers with "physical insight" into their models



As they are so important, the special sets of inputs which just get "amplified" by a system are given a name and are a major focus of engineering maths

Terminology: The solutions to $Ax = kx$ are called *eigensolutions*, x is an *eigenvector* and the corresponding value of k , the *eigenvalue*

Given our time constraints, we will focus on when they exist, what are their properties and how can they be exploited. We will not explore how to find them in practice and also will gloss over some issues as that would require a dedicated 20 credit module on its own! See Self Study Sheet #3 for some suggest further learning opportunities.

We have motivated ourselves as engineers to explore eigensolutions, but now let's be selfishly computational mathematicians and ask

What's in for us?

Eigen-solutions



Matrix multiplication - EASY

Matrix solving - HARD

Example: We have a system represented by the matrix problem $Ax = b$. A is a *big* matrix. If the desired output, b , is chosen to be an *eigenvector* of the system with eigenvalue 2, what input x should be applied to obtain it?

Solution:

The definition of an eigensolution is $Ax = kx$.

We are told that $Ab = 2b$.

So if we choose x to be $0.5b$

$Ax = A(0.5b) = 0.5(Ab) = 0.5(2b) = b$ as required

With eigenvectors

Matrix multiplication - TRIVIALY EASY

Matrix solving - TRIVIALY EASY

Although we will not discuss *how* to find the eigensolutions in general, we ought to at least convince ourselves that such things can exist

Example: Consider the small matrix problem
$$\begin{pmatrix} y_0 \\ y_1 \end{pmatrix} = \begin{pmatrix} 19 & -2 \\ -2 & 16 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix}$$

if $\begin{pmatrix} x_0 \\ x_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ then $\begin{pmatrix} y_0 \\ y_1 \end{pmatrix} = \begin{pmatrix} 15 \\ 30 \end{pmatrix} = 15 \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

if $\begin{pmatrix} x_0 \\ x_1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ then $\begin{pmatrix} y_0 \\ y_1 \end{pmatrix} = \begin{pmatrix} 40 \\ -20 \end{pmatrix} = 20 \begin{pmatrix} 2 \\ -1 \end{pmatrix}$

In both case $\begin{pmatrix} 19 & -2 \\ -2 & 16 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} = \text{scaling factor} \times \begin{pmatrix} x_0 \\ x_1 \end{pmatrix}$

$$\mathbf{M} \mathbf{V}_{in} = k \mathbf{V}_{in}$$

The eigensolutions are:

Eigenvector = $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, Eigenvalue = 15

Eigenvector = $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$, Eigenvalue = 20

Note how in the control/bridge/optical fibre I may have phrased it to imply that there is only one possible special case of $\mathbf{x}$ such that $A\mathbf{x} = k\mathbf{x}$

However, this last example revealed that there are 2 eigensolutions

In general, there will either exist

- No eigensolutions
- *or* A finite number of eigensolutions
- *or* An infinite number of eigensolutions

Wasn't that informative!

Multiple Eigen-solutions

How to visualise multiple eigensolutions

Eigensolutions are can often be analogous to *resonances* of a system. Each eigensolution corresponds to a different resonant case

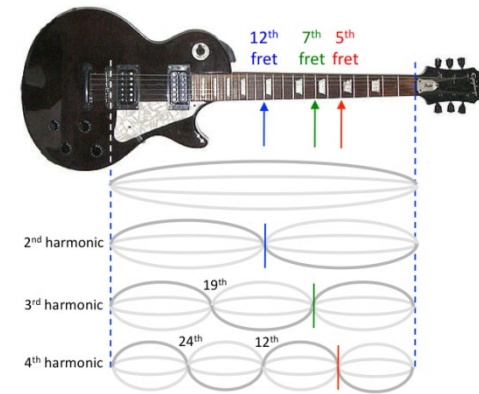
The bridge examples - similar mechanical resonances

In the light down a fibre case, each eigensolution corresponds to a different pattern of light that *doesn't get distorted in shape* as it travels down the fibre

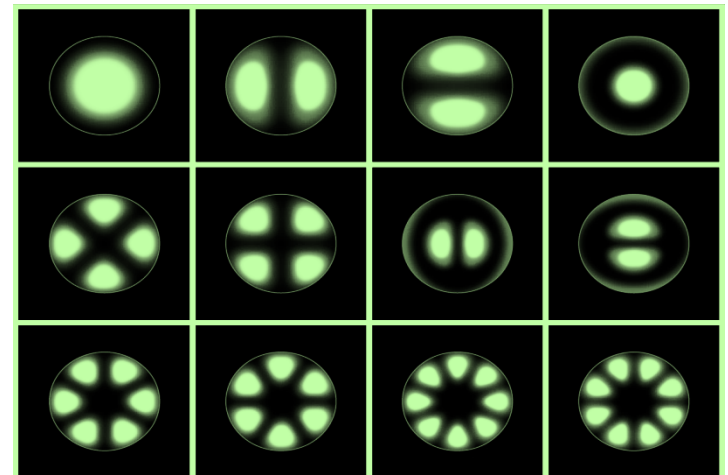


Harmonics on a Guitar

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http://68.media.tumblr.com/tumblr_lh96t0EvPA1qbtjkw01_1280.png



<https://phys.org/news/2017-07-optical-fiber-properties.html>



Surely the chances of my inputs being an eigenvector are negligible, except for a few special, albeit important, cases!

Multiple Eigen-solutions



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We are heading for the mega-important *It can be shown result* here.

However, rather than just quote it let us least see if we can first give ourselves an inclination to believe it based on what we know

Word association game:

Guitar string **HARMONICS** are **eigensolutions**

Fourier series representation of waveforms says that **ANY** (periodic) waveform can be represented as a suitable **sum of** **HARMONICS** of the fundamental frequency

Matrices can be used to represent sampled time waveforms (DFT)

ANY solution of a matrix problem can be represented by a suitable **sum of** its **eigensolutions**

ANY solution, x and b , of a matrix problem $Ax = b$ can each be represented by a suitable **superposition of** the eigensolutions of A

Except when it can't!

Another helpful and informative statement!

Strictly speaking it's more complicated than this if you really want to be a super precise mathematician and we will lightly touch on the issues as we proceed here

Typically, an engineer knows that their matrices have certain properties (e.g. all nodal analysis matrices are symmetric) that allows them to confidently and correctly use the above result and that is what we will assume for the moment

How does this result revolutionise electrical and electronic engineering analysis?

You should know - you have been using this result for two years now!

ANY solution, x and b , of a matrix problem $Ax = b$ can each be represented by a suitable **superposition of** the eigensolutions of A

This is the rigorous mathematic justification that allows

Phasor Analysis

Fourier Series; Fourier, Laplace and Z Transforms

Spectrum analysers and communications systems analysis

It is the basis of, or at least a major design tool for

Advanced control design

Advanced communications systems

Advanced Signal processing

Power network analysis

Types of data coding error correction used in your phones

Modern spread spectrum techniques (5th Gen. etc)

Radio/micro/light wave propagation

Large scale numerical simulations and predictions of physical phenomena ...

Got to be worth at least a quick look then?

Example: Express in matrix notation the fact that a column vector, $\mathbf{y}$, can be represented as a superposition of a set of eigensolutions of a matrix $\mathbf{A}$

Solution: We will assume there are N eigensolutions and label each as $\mathbf{x}_i$ with a corresponding eigenvalue, k_i

So $\mathbf{A}\mathbf{x}_i = k_i\mathbf{x}_i$ is the definition of each eigensolution

The superposition will be of the form

$$\mathbf{y} = \sum_{i=0}^{N-1} \mathbf{x}_i c_i$$

where c_i is the "amount of", better said as coefficient of, the i^{th} eigenvector included in the superposition. We don't know to get these yet - but they exist.

It can be helpful initially that we expand out the notation to emphasis the nature of each part

$$\begin{pmatrix} y_0 \\ y_1 \\ \dots \end{pmatrix} = \begin{pmatrix} x_{0,0} \\ x_{0,1} \\ \dots \end{pmatrix} c_0 + \begin{pmatrix} x_{1,0} \\ x_{1,1} \\ \dots \end{pmatrix} c_1 + \begin{pmatrix} x_{2,0} \\ x_{2,1} \\ \dots \end{pmatrix} c_2 \dots \begin{pmatrix} x_{N-1,0} \\ x_{N-1,1} \\ \dots \end{pmatrix} c_{N-1}$$

Multiple Eigen-solutions



$$\begin{pmatrix} y_0 \\ y_1 \\ \dots \end{pmatrix} = \begin{pmatrix} x_{0,0} \\ x_{0,1} \\ \dots \end{pmatrix} c_0 + \begin{pmatrix} x_{1,0} \\ x_{1,1} \\ \dots \end{pmatrix} c_1 + \begin{pmatrix} x_{2,0} \\ x_{2,1} \\ \dots \end{pmatrix} c_2 \dots \begin{pmatrix} x_{N-1,0} \\ x_{N-1,1} \\ \dots \end{pmatrix} c_{N-1}$$

$$\mathbf{y} = \sum_{i=0}^{N-1} \mathbf{x}_i c_i$$

Now we will put the into the final conventional format

$$\begin{pmatrix} y_0 \\ y_1 \\ \dots \end{pmatrix} = \begin{pmatrix} x_{0,0} & x_{1,0} & x_{2,0} & \dots & x_{N-1,0} \\ x_{0,1} & x_{1,1} & x_{2,1} & \dots & x_{N-1,1} \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ \dots \\ c_{N-1} \end{pmatrix}$$

I doubt this is obvious the first time you see it. Check it multiplies out correctly - note how each coefficient only combines with the column of values it ought to

$$\begin{pmatrix} y_0 \\ y_1 \\ \dots \end{pmatrix} = \begin{pmatrix} x_{0,0} & \mathbf{x_{1,0}} & x_{2,0} & \dots & x_{N-1,0} \\ x_{0,1} & \mathbf{x_{1,1}} & x_{2,1} & \dots & x_{N-1,1} \\ \dots & \mathbf{\dots} & \dots & \dots & \dots \end{pmatrix} \begin{pmatrix} c_0 \\ \mathbf{c_1} \\ c_2 \\ \dots \\ c_{N-1} \end{pmatrix}$$

$$\mathbf{y} = \mathbf{Xc}$$

$$\mathbf{y} = \mathbf{X} \mathbf{c}$$

Example: Demonstrate that using the previous representation makes solving $Ax = b$ for x given A and b relatively trivial!

Solution: Both the unknown x being sought and the known b can be represented as above

$x = Xc$ $b = Xd$ where c is to be found and d is known

$AXc = Xd$ First time out it is clearer to rewrite this as

$$A \sum_{i=0}^{N-1} x_i c_i = \sum_{i=0}^{N-1} x_i d_i$$

$$\sum_{i=0}^{N-1} Ax_i c_i = \sum_{i=0}^{N-1} x_i d_i \quad \text{use the def'n of eigensolutions} \quad \sum_{i=0}^{N-1} k_i x_i c_i = \sum_{i=0}^{N-1} x_i d_i$$

Compare the amounts of x_i on each side of the equation, $d_i = k_i c_i$, hence $c_i = \frac{d_i}{k_i}$

$x = Xc$ with the elements of the output coefficients, c , being $c_i = \frac{d_i}{k_i}$ given the input is known to be $b = Xd$

Multiple Eigen-solutions



Review the previous example:

*If you can identify **all** the eigensolutions of **A***

Really hard work!

and you have a way of working how much of each one is required to write the input as

$$\sum_{i=0}^{N-1} x_i d_i$$

But we don't!

then it is trivially easy to solve $Ax = b$

Doesn't it rather sound like we are just moving the "effort from one place to another!"

I cannot deny that!

But its not quite as bad as we might think due to ...

It can be shown that: The matrix of eigenvectors, $\mathbf{X}$, above is an orthogonal matrix

Except when it isn't - sorry!

I am comfortable to say, that for many engineeringly important matrices, the above *is* true. Specifically, if the matrix $\mathbf{A}$ has real elements and is symmetric, i.e. $\mathbf{A}^T = \mathbf{A}$

Note how nodal analysis matrices for example fall into this category

Example: Show how the orthogonality allows us to easily evaluate the required coefficient values in

$$\mathbf{b} = \mathbf{X}\mathbf{d} = \sum_{i=0}^{N-1} \mathbf{x}_i d_i$$

Solution:

$$\mathbf{b} = \mathbf{X}\mathbf{d}, \quad \mathbf{X}^T \mathbf{b} = \mathbf{X}^T \mathbf{X} \mathbf{d} \quad \mathbf{X}^T \mathbf{b} = \mathbf{d}$$

So let's now use this in the previous problem

Example: Demonstrate that using the previous representation makes solving $Ax = b$ for x given A and b relatively trivial!

Solution: $x = Xc$ $b = Xd$ and we showed that $c_i = \frac{d_i}{k_i}$

$$x = Xc = X \begin{pmatrix} \frac{1}{k_0} & & \\ & \frac{1}{k_1} & \\ & & \dots \end{pmatrix} d = X \begin{pmatrix} \frac{1}{k_0} & & \\ & \frac{1}{k_1} & \\ & & \dots \end{pmatrix} X^T b$$

$$x = X \begin{pmatrix} \frac{1}{k_0} & & \\ & \frac{1}{k_1} & \\ & & \dots \end{pmatrix} X^T b$$



Surely the chances of my inputs being an eigenvector are negligible, except for a few special, albeit important, cases!

It doesn't matter, we can express the solution for any input in terms of the eigensolutions

This *is* why Fourier Series, DFTs etc are legitimate representations



We have done a lot in different ways - can we start to make connections between the different aspects?

Example: Consider how the eigensolution representation used to solve $Ax = b$ reveals the same "solvability" information as the QR decomposition

Solution: when we discussed QR for solving $Ax = b$ we said ...when we solve the triangular problem when one or more diagonal elements is 0 we cannot proceed to solve with certainty

$$\begin{pmatrix} * & * & * \\ 0 & \mathbf{0} & * \\ 0 & 0 & * \end{pmatrix} x = \begin{pmatrix} * \\ * \\ * \end{pmatrix}$$

With the eigensolution approach, don't we have something similar?

$$x = X \begin{pmatrix} k_0^{-1} & & \\ & k_1^{-1} & \\ & & \dots \end{pmatrix} X^T b$$

If one or more of the eigenvalues, k_i , more diagonal elements were 0? We cannot proceed with certainty

- Knowledge of all the eigenvalues (we properly say knowledge of the *spectrum*) of the matrix A reveals whether $Ax = b$ can be uniquely solved
- **It can be shown that:** The diagonal elements of the R matrix in a **QR** decomposition of A are the eigenvalues of A

Example: Consider how the eigensolution representation used to solve $Ax = b$ relates to the concept of the inverse matrix A^{-1}

Solution: The definition of the inverse suggests $x = A^{-1} b$ and we have obtained

$$x = X \begin{pmatrix} k_0^{-1} & & \\ & k_1^{-1} & \\ & & \dots \end{pmatrix} X^T b \quad \text{so} \quad A^{-1} = X \begin{pmatrix} k_0^{-1} & & \\ & k_1^{-1} & \\ & & \dots \end{pmatrix} X^T$$

A way of evaluating that inverse matrix at last!

Note that $A^{-1} A = I$ leads us to propose that

$$A = X \begin{pmatrix} k_0 & & \\ & k_1 & \\ & & \dots \end{pmatrix} X^T \quad \text{The eigen decomposition theorem, valid if } A \dots$$

Does it work?

$$A^{-1} A = X \begin{pmatrix} k_0 & & \\ & k_1 & \\ & & \dots \end{pmatrix} X^T X \begin{pmatrix} k_0^{-1} & & \\ & k_1^{-1} & \\ & & \dots \end{pmatrix} X^T = X X^T = I$$



Reflection: We were told in 1st year that a set of simultaneous equations could only be solved if the *determinant* of the matrix is not zero. How does this relate to this discussion?

In 1st year you are given a pure recipe for evaluating the determinant of a *square* matrix, $|A|$ (or $\det(A)$).

We are saying that a matrix problem can only be solved uniquely as long as all the eigenvalues are non zero.

It can be shown that: $|A| = k_0 k_1 k_2 \dots k_{N-1}$, so that if one or more eigenvalues are zero, *so is the determinant*. We may like to say that the determinant gives an overall measure of solvability whereas knowledge of the eigenvalues (which is much harder to obtain) provides more detail about *in what way it can't be solved* - *which means we can try to get around the difficulty*.

The Positive View of Zero eigenvalues



We have really been negative about 0 eigenvalues, e.g. can't uniquely solve $Ax = b$

But sometimes we formulate matrix problems to express an error that we really would like to be 0!

Example: A flight control system measures a number of *error* values; maybe the error in height, error in direction ... We group these values in an error column, $\mathbf{e}$. These errors are related to the control inputs, $\mathbf{x}$, as $A\mathbf{x} = \mathbf{e}$. Is it possible to find a set of inputs to get zero errors and if so, what should they be?

Solution: We must be seeking $\mathbf{x}$, such that $A\mathbf{x} = \mathbf{0}$

Consider this as $A\mathbf{x} = 0\mathbf{x}$, so we are looking for the eigenvector associated with an eigenvalue of 0 - if one exists

Is any eigenvalue 0? Well $|A| = k_0 k_1 k_2 \dots k_{N-1}$ so there only exists an error free solution *if and only if*, the determinant of A is zero.

If this is the case, we could use one of the standard eigensolution finding routines to get the associated eigen vector.



Example: We can actually use this logic to identify general eigenvalues: This is the standard "text book" way to find all eigensolutions of very small matrices

Solution: We seek k and $\mathbf{x}$ such $A\mathbf{x} = k\mathbf{x}$

Express this as $(A - kI)\mathbf{x} = 0$ which can only have a solution if the determinant of the matrix $(A - kI)$ is zero.

From the 1st year, for a 2 by 2 case, the determinant of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is $ad - bc$

$$|A - kI| = \begin{vmatrix} A_{00} - k & A_{01} \\ A_{10} & A_{11} - k \end{vmatrix} = (A_{00} - k)(A_{11} - k) - A_{01}A_{10}$$

So, $A\mathbf{x} = k\mathbf{x}$ has eigenvectors, k , which satisfy $(A_{00} - k)(A_{11} - k) - A_{01}A_{10} = 0$

Now let's do a numerical example to cement this

Example: Identify all the eigensolutions of $A = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$

Solution:

$$|A - kI| = \begin{vmatrix} 1-k & 2 \\ 2 & 5-k \end{vmatrix} = (1-k)(5-k) - 2*2$$

So, $Ax = kx$ has eigenvectors, k , which satisfy $k^2 - 6k + 1 = 0$

$$k = \frac{6 \pm \sqrt{36 - 4}}{2} \quad \text{hence, } k_0 = 0.172 \text{ and } k_1 = 5.83$$

$$\text{To find the eigenvectors use } (A - k_i I) \begin{pmatrix} x_{i0} \\ x_{i1} \end{pmatrix} = \begin{pmatrix} 1-k_i & 2 \\ 2 & 5-k_i \end{pmatrix} \begin{pmatrix} x_{i0} \\ x_{i1} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We can always choose the 1st element, x_{i0} , to be 1 leaving

$$(1 - k_i) + 2x_{i1} = 0$$

$$x_0 = \begin{pmatrix} 1 \\ -0.41 \end{pmatrix} \text{ with } k_0 = 0.172$$

$$x_1 = \begin{pmatrix} 1 \\ 2.41 \end{pmatrix} \text{ with } k_1 = 5.83$$

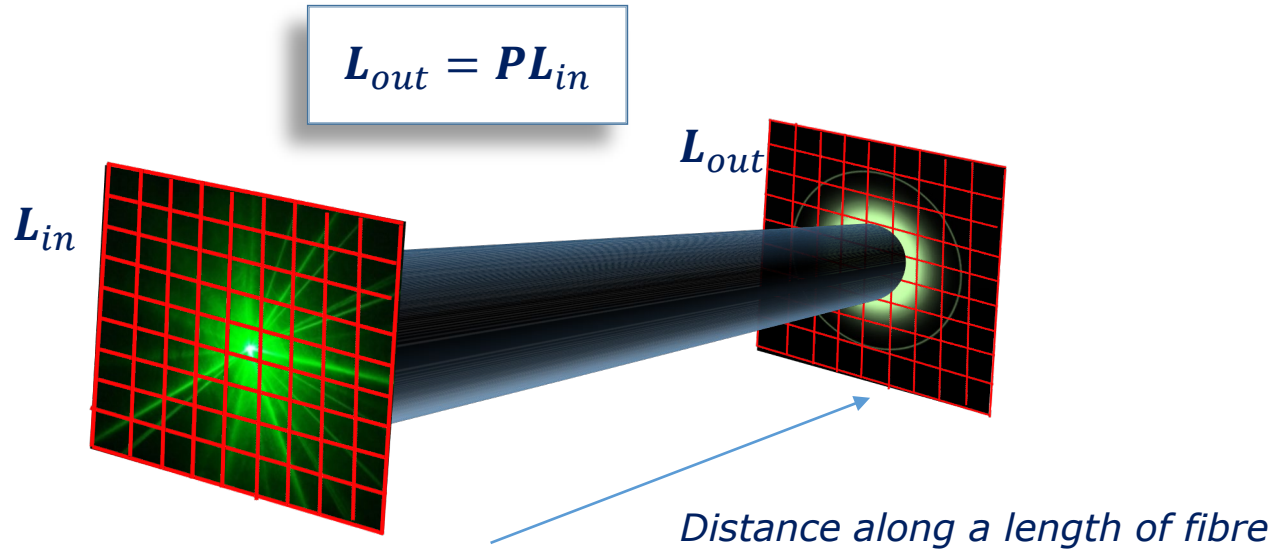


**Let us close by revisiting our case studies
with the power of eigensolutions**

Eigen Decomposition Theorem applied to fibre optics



Example: Show how with careful design, an optical fibre can be made so that after a short length, the light pattern settles to a stable pattern



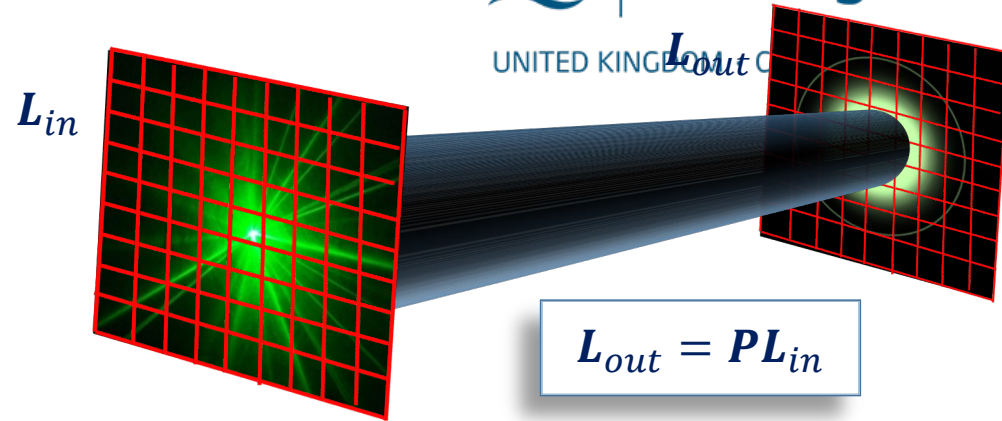
The input light pattern can be represented as a superposition of eigensolutions of the propagation matrix $\mathbf{P}$

$$\mathbf{L}_{out} = \mathbf{X} \begin{pmatrix} k_0 & & \\ & k_1 & \\ & & \dots \end{pmatrix} \mathbf{X}^T \mathbf{L}_{in} \quad \text{where } \mathbf{X} \text{ is the matrix of eigenvectors of } \mathbf{P} \text{ and } k_i$$

are the corresponding eigenvalues

Eigen Decomposition Theorem applied to fibre optics

$$L_{out} = X \begin{pmatrix} k_0 & & \\ & k_1 & \\ & & \dots \end{pmatrix} X^T L_{in}$$

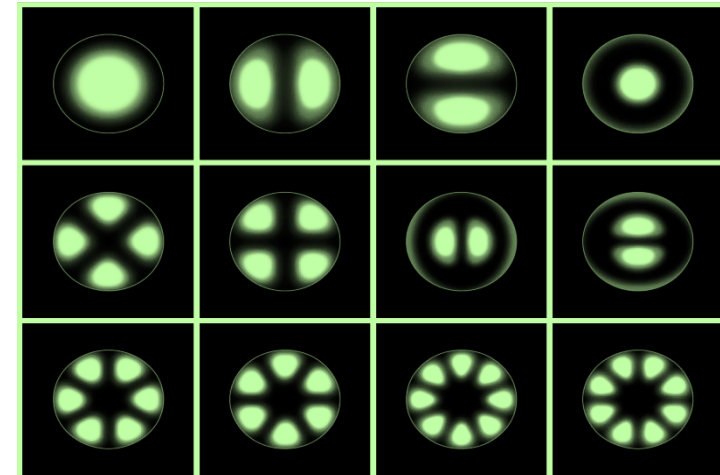


In practice, most of the values of k_i will be really small and only one, i.e. k_0 , is ~ 1

$$L_{out} \sim X \begin{pmatrix} 1 & & \\ & 0 & \\ & & \dots \end{pmatrix} X^T L_{in} \sim X_0 \text{ times a constant}$$

The fibre behaves as an optical pattern filter?

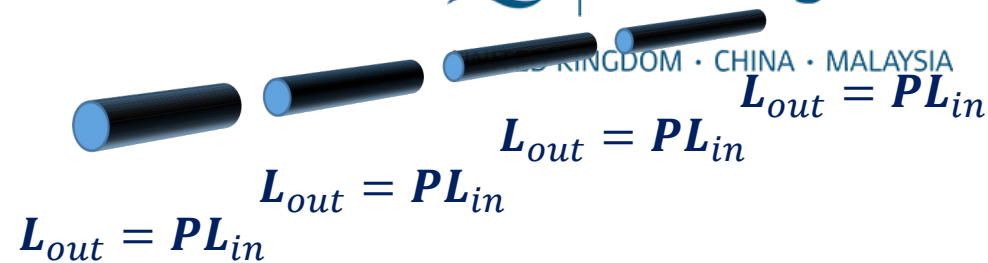
For information, such a situation is known as *single mode* propagation and this is exactly the requirement for long distances fibres



Eigen Decomposition Theorem applied to fibre optics



$$\mathbf{L}_{out} = \mathbf{X} \begin{pmatrix} k_0 & & \\ & k_1 & \\ & & \dots \end{pmatrix} \mathbf{X}^T \mathbf{L}_{in}$$



An interesting development of this is to consider a cascade of small lengths of fibre, the propagation along *each of which* is modelled by $\mathbf{L}_{out} = \mathbf{P}\mathbf{L}_{in}$

Using our cascaded circuit picture of matrix multiplication, the *overall* transfer of light is give by $\mathbf{L}_{out} = \mathbf{P}\mathbf{P}\mathbf{P} \dots \mathbf{P}\mathbf{L}_{in}$. I suppose we write this as a power function $\mathbf{L}_{out} = \mathbf{P}^M \mathbf{L}_{in}$ as a shorthand

What happens if we use the eigendecomposition theorem, above

$\mathbf{L}_{out} = \mathbf{P}\mathbf{P}\mathbf{P} \dots \mathbf{P}\mathbf{L}_{in} = \mathbf{X}\mathbf{k}\mathbf{X}^T \mathbf{X}\mathbf{k}\mathbf{X}^T \mathbf{X}\mathbf{k}\mathbf{X}^T \mathbf{X}\mathbf{k}\mathbf{X}^T \dots \mathbf{X}\mathbf{k}\mathbf{X}^T \mathbf{L}_{in}$ and use the fact that $\mathbf{X}^T \mathbf{X} = \mathbf{I}$

$$\mathbf{L}_{out} = \mathbf{P}^M \mathbf{L}_{in} = \mathbf{X} \begin{pmatrix} k_o^M & & \\ & k_1^M & \\ & & \dots \end{pmatrix} \mathbf{X}^T \mathbf{L}_{in}$$

Eigen Decomposition Theorem applied to fibre optics



It can be shown that: If the matrix A has the eigen representation $A = XkX^T$ where X contains the eigenvectors of A and k is a diagonal matrix whose elements are the eigenvalues of A , then

The eigenvectors of A^M are the same as those of A and the eigenvalues are k_i^M ,

$$\text{hence } A^M = X \begin{pmatrix} k_o^M & & \\ & k_1^M & \\ & & \dots \end{pmatrix} X^T$$

This result has an amazing generalisation

It can be shown that:... as above, and $f()$ is any function, then

The eigenvectors of $f(A)$ are the same as those of A and the eigenvalues are $f(k_o)$

Example: Consider the exponential of a matrix - !!!!!!!! e^A , then

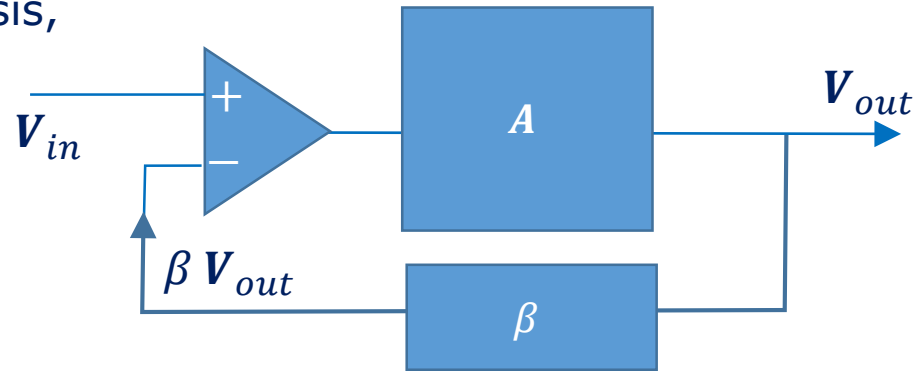
$$e^A = X \begin{pmatrix} e^{k_o} & & \\ & e^{k_1} & \\ & & \dots \end{pmatrix} X^T$$

Not just maths games but np time to explore further!

Eigen Decomposition Theorem applied to control stability



Example: For a feedback system analysis, we use column vectors containing time samples of the waveforms as before, a system modelled by a matrix $\mathbf{A}$ whose eigendecomposition is available, and simple scalar feedback. $(\mathbf{I} + \beta \mathbf{A})\mathbf{V}_{out} = \mathbf{A} \mathbf{V}_{in}$



Determine the eigendecomposition of the overall system with feedback

Solution: We are given the data to construct $\mathbf{A} = \mathbf{X} \mathbf{k} \mathbf{X}^T$

Standard analysis gives $(\mathbf{I} + \beta \mathbf{A})\mathbf{V}_{out} = \mathbf{A} \mathbf{V}_{in}$

$\mathbf{I} + \beta \mathbf{A}$ is a "function of a matrix" so $\mathbf{I} + \beta \mathbf{A} = \mathbf{X} \begin{pmatrix} 1 + \beta k_0 & & \\ & 1 + \beta k_1 & \\ & & \dots \end{pmatrix} \mathbf{X}^T$

$$\mathbf{V}_{out} = \mathbf{X} \begin{pmatrix} \frac{k_0}{1 + \beta k_0} & & \\ & \frac{k_1}{1 + \beta k_1} & \\ & & \dots \end{pmatrix} \mathbf{X}^T \mathbf{V}_{in}$$

The system goes unstable if any eigenvalue satisfies $1 + \beta k_i = 0$

Not bad - stability analysis of multi input/output feedback control loop!